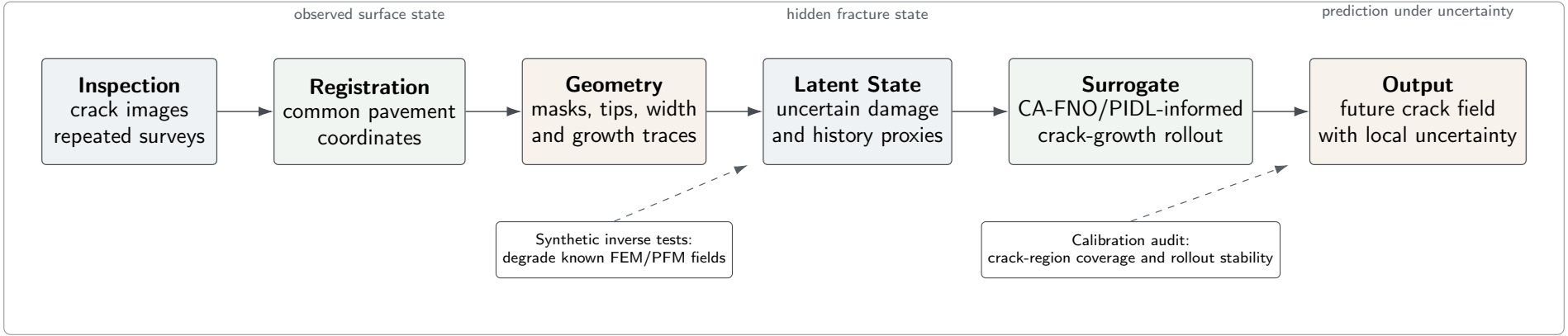
observed surface state
hidden fracture state
prediction under uncertainty
Inspection
crack images
repeated surveys
Registration
common pavement
coordinates
Geometry
masks, tips, width
and growth traces
Latent State
uncertain damage
and history proxies
Surrogate
CA-FNO/PIDL-informed
crack-growth rollout
Output
future crack field
with local uncertainty
Synthetic inverse tests:
degrade known FEM/PFM fields
Calibration audit:
crack-region coverage and rollout stability